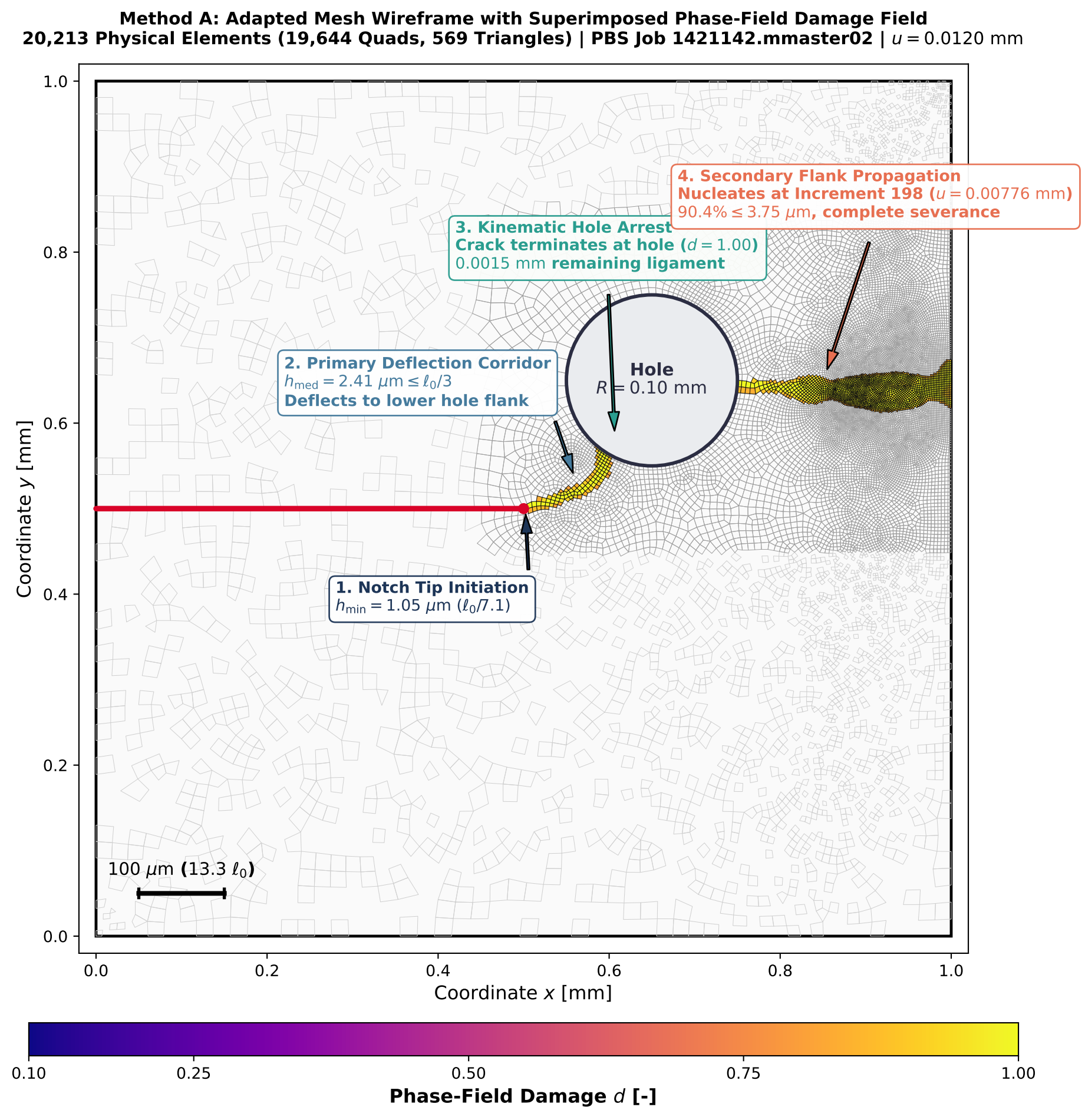

Method A: Adapted Mesh Wireframe with Superimposed Phase-Field Damage Field
20,213 Physical Elements (19,644 Quads, 569 Triangles) | PBS Job 1421142.mmaster02 | u = 0.0120 mm
4. Secondary Flank Propagation
Nucleates at Increment 198 (u = 0.00776 mm)
90.4% ≤ 3.75 μm, complete severance
3. Kinematic Hole Arrest
Crack terminates at hole (d = 1.00)
0.0015 mm remaining ligament
Hole
R = 0.10 mm
2. Primary Deflection Corridor
h_med = 2.41 μm ≤ ℓ0/3
Deflects to lower hole flank
1. Notch Tip Initiation
h_min = 1.05 μm (ℓ0/7.1)
100 μm (13.3 ℓ0)
Coordinate y [mm]
Coordinate x [mm]
0.0
0.2
0.4
0.6
0.8
1.0
0.10
0.25
0.50
0.75
1.00
Phase-Field Damage d [-]